

Ê PHASOR TEMPLATES (Air: $\epsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega \sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad (\text{incident wave, Med 1})$$

$$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad (\text{reflected wave, Med 1})$$

$$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)

Total in medium 1: $\mathbf{E}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.

	Normal $\perp$ pol ($\theta_i = 0$)	$\parallel$ pol
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\parallel} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
$R+T$ total	$T = 1 - R$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; nonmag. $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

BOUNDARY

η_1, η_2 — imp. (Ω)
 Γ, τ, R, T — coeffs (unitless)
 $\tau = 1 + \Gamma$; $R = |\Gamma|^2$
 $k_i = \omega \sqrt{\epsilon_{r,i}} / c$ (rad/m)
 α_2 (Np/m), β_2 (rad/m): lossy
 $z=0$ — boundary plane (m)
 $\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
 $n_i = \sqrt{\epsilon_{r,i}}$ — index of refr.
 t_i — slab thickness (m)
 d — lateral disp. (m)
plane of inc. = plane of $\hat{k}$ +normal
 $\perp$: $\mathbf{E} \perp$ this plane
 $\parallel$: $\mathbf{E} \parallel$ this plane

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r,1}}{\epsilon_{r,2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs* $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

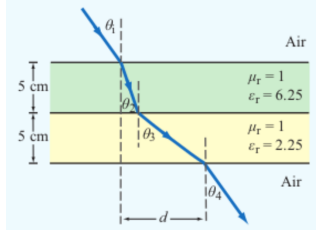
$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i , refracted angle inside slab i

Slab-indexed $\theta_i =$ angle inside slab i ; $t_i =$ slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	$\sigma/\omega\epsilon$	η_c, α, β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact $\alpha=0, \beta = \omega\sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$
Quasi-cond.	10^{-2} – 10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$

For Γ/τ : drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .

Low-loss quick params $\sigma/(\omega\epsilon) \ll 1, \mu_r = 1$

$$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$$

$$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$

TM mode $H_z = 0, E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE $_{mn}$ vs TM $_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” $\rightarrow$ TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).

3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE $_{10}$ exists but TM $_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\hat{E}$ in V/m, $\hat{H}$ in A/m.

TE mode *generator: $\hat{H}_z$*

$$\hat{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

$$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

$$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TE}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TE}}$$

TM mode *generator: $\hat{E}_z$*

$$\hat{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

$$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TM}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference: $\hat{E}_x = E_0 e^{-j\beta z}, \hat{H}_y = \hat{E}_x/\eta, f_c$ N/A, $k = \omega\sqrt{\mu\epsilon}$.

CUTOFF FREQUENCY

f_{mn} (a.k.a. f_c for that mode) m, n integers ≥ 0 (not both zero)

$$f_c \equiv f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs ($a > b$)

Mode	f_{mn}
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE $_{10}$	$f_{10} = u_{p0}/(2a)$ (dominant)
TE $_{01}$	$f_{01} = u_{p0}/(2b)$
TE $_{20}$	$f_{20} = u_{p0}/a = 2f_{10}$
TE $_{11}, \text{TM}_{11}$	$f_{11} = \frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$. In $Z_{\text{TE}}/Z_{\text{TM}}$ and u_g , $f_c = f_{mn}$ of the excited mode.

ϵ_r FROM DISPERSION

Given ω, β , mode (m, n) *result is unitless*

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs: ω (rad/s), β (rad/m), a, b (m), $c = 3 \times 10^8$ (m/s), m, n (unitless).

WAVE IMPEDANCE & H_y

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2} \text{ (}\Omega\text{)}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)} \quad H_y = \frac{E_x}{Z_{\text{TE}}} \text{ (A/m, TE mode)}$$

$Z_{\text{TE}} > \eta$ always (denominator < 1); $Z_{\text{TM}} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}^2}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower $u_g \Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$ — bulk phase vel.
 $u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$
 $u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$
 $u_p u_g = u_{p0}^2$
 $t = L/u_g$ — travel time
 $c = 3 \times 10^8$ m/s

POWER & UNITS

$\%P_r, \%P_t$ — pwr refl./trans. (unitless)
 $\%P_r = 100 |\Gamma|^2$
 $\%P_t = 100 |\tau|^2 \eta_1/\eta_2$
 $\%P_r + \%P_t = 100$
 σ (S/m), ϵ (F/m), μ (H/m)
 $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
 $\mu_0 = 4\pi \times 10^{-7}$ H/m

WAVEGUIDE

a, b — wide, narrow dim. (m);
 $a > b$
 m, n — mode indices (int.)
 f_{mn} — cutoff (Hz)
 β — prop. const (rad/m)
 Z_{TE}/TM — wave imp. (Ω)
 E_x — E-field comp. (V/m)
 H_y — H-field comp. (A/m)
 $\eta = \eta_0/\sqrt{\epsilon_r}$

SNELL / OBLIQUE